

EMPLOYEE ATTRITION & WORKFORCE ANALYTICS DASHBOARD

Project Documentation — Power BI

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Tool Used:
Microsoft Power BI

Project:
ShadowFox Data Analyst Internship — Advanced Level

Tool Used: Microsoft Power BI (Report View, Data Model, DAX)
Dataset: HR_Data (Employee-Level Records)
Theme: Dark + Neon Red

*In one line: raw employee records, transformed into an interactive dashboard
that shows where attrition is concentrated and what HR can do about it.*

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Executive Summary

This project delivers an interactive Employee Attrition & Workforce Analytics Dashboard built in Microsoft Power BI, using employee-level HR records (HR_Data) as the source. The dashboard was designed as a decision-support tool for HR stakeholders rather than a static report: five KPI cards, three interactive slicers, and seven attrition-rate visuals let a user move from an overall attrition rate of 16.12% down to the specific departments, roles, tenure bands, age groups, income bands, and satisfaction levels driving that number.

The analysis surfaces four consistent, data-backed risk areas — the Sales Representative role, overtime, early tenure and early career, and the lowest income band — and translates each into a targeted, actionable HR recommendation. A dark-and-neon-red visual theme was applied deliberately to reinforce the dashboard's focus on risk and attrition, and the finished build was tested across its slicers, DAX logic, and formatting before being considered complete.

Project Overview

The project was built end-to-end in Microsoft Power BI, using its Report View, Data/Model View, DAX (calculated columns and measures), interactive slicers, and dashboard formatting and layout tools. A dedicated Measures table was created to keep every DAX measure organized in one place rather than scattered across the model.

The dashboard is built on the HR_Data table, which holds employee-level information including Age, Gender, Department, Job Role, Attrition (Yes/No), Monthly Income, OverTime, Years at Company, Job Satisfaction, and a number of other employee attributes. These fields form the backbone of every calculation and visual in the dashboard.

Problem Statement

Employee attrition carries a real, ongoing cost — in hiring, onboarding, lost productivity, and institutional knowledge — but that cost is only manageable if HR can see where it is concentrated. Raw, employee-level HR data does not answer that question on its own: without a way to break attrition down by department, role, tenure, income, age, and satisfaction, HR is left reacting to attrition after the fact rather than identifying and addressing the specific segments driving it. This project addresses that gap by turning raw employee records into an interactive tool that HR can use day to day.

Objectives

The dashboard was built to answer the following questions for HR:

- How many employees are in the workforce, and how many have left
- What the overall attrition rate looks like
- Which departments have higher attrition than others
- Which job roles are the highest risk
- Whether overtime is associated with higher attrition

- Whether income level is associated with attrition
- Which age groups are losing the most people
- Which tenure groups are the most at risk
- Whether job satisfaction correlates with attrition

In short, the objective was to convert employee-level data into an interactive HR decision-support dashboard — not a static report.

Dataset Description

The dashboard is built on a single primary table, HR_Data, containing employee-level records with the following key fields:

- Age
- Gender
- Department
- Job Role
- Attrition (Yes/No)
- Monthly Income
- OverTime
- Years at Company
- Job Satisfaction

...along with a number of other employee attributes retained in the source data.

Data Preparation & Transformation

Column creation was kept deliberately minimal — only calculated columns that would genuinely help HR interpret the data were added.

Age Group

Employees were grouped into five bands — Under 25, 25–34, 35–44, 45–54, and 55+ — rather than shown by individual age, making the data far easier for a stakeholder to scan. Because text labels like these don't sort correctly on their own (Power BI would alphabetize them), an Age Group Sort helper column was added so the bands display in the correct order on every chart.

Income Group

Monthly Income was grouped into five bands: Below ₹3K, ₹3K–₹5K, ₹5K–₹7K, ₹7K–₹10K, and ₹10K+. As with Age Group, an Income Group Sort column was added so the bands appear in true income order rather than alphabetical order.

Tenure Group

YearsAtCompany was converted into five lifecycle stages — 0–1 Year, 2–3 Years, 4–5 Years, 6–10 Years, and 10+ Years — enabling a view of retention risk across the employee lifecycle instead of individual year counts. A sort column again keeps these in the correct chronological order.

Job Satisfaction Level

The original dataset held only numeric scores (1 through 4). A readable label column was created: 1 - Low, 2 - Medium, 3 - High, 4 - Very High. A separate sort column was not needed here, since the numeric prefix already keeps the labels in the correct order.

Data Model & DAX Measures

Rather than building calculations separately inside every visual, reusable DAX measures were created. This keeps the model clean and, more importantly, means every calculation automatically responds to whatever slicers or filters are active on the page.

Total Employees

```
Total Employees =  
COUNTROWS(HR_Data)
```

Counts every row in HR_Data — the base employee headcount that every other measure builds on.

Employees Left

```
Employees Left =  
CALCULATE(  
    COUNTROWS(HR_Data),  
    HR_Data[Attrition] = "Yes"  
)
```

Counts only the employees whose Attrition field is "Yes".

Attrition Rate

```
Attrition Rate =  
DIVIDE(  
    [Employees Left],  
    [Total Employees],  
    0  
)
```

Divides Employees Left by Total Employees, with a safe 0 fallback if a filter context returns zero employees.

Average Monthly Income

```
Avg Monthly Income =  
AVERAGE(HR_Data[MonthlyIncome])
```

Average Years at Company

```
Avg Years at Company =  
AVERAGE(HR_Data[YearsAtCompany])
```

Filter Context

This is one of the ideas the whole dashboard leans on. The base attrition-rate logic is simply Employees Left divided by Total Employees. The moment Department is placed on a chart, Power BI automatically re-evaluates that same measure separately for every department in the filter context — no separate formula is required for each one. For example, breaking the same measure down by department gives:

Department	Attrition Rate
Sales	20.63%
Human Resources	19.05%
Research & Development	13.84%

The same principle applies when Department is swapped for Job Role, Overtime, Income Group, Tenure Group, Age Group, or Job Satisfaction — which is why one small set of measures is enough to power the entire dashboard.

Dashboard Design

Theme: Dark + Neon Red

A dark background with neon red accents was chosen specifically to build a strong visual hierarchy around risk and attrition, while keeping everything easy to read.

Element	Hex Code
Background	#050608
Visual / Card Background	#0D0F12
Primary Red	#FF1744
Highlight Red	#FF2B55
White (text)	#FFFFFF
Light Gray (secondary text)	#D1D5DB
Muted Gray (labels)	#9CA3AF
Border	#25282D

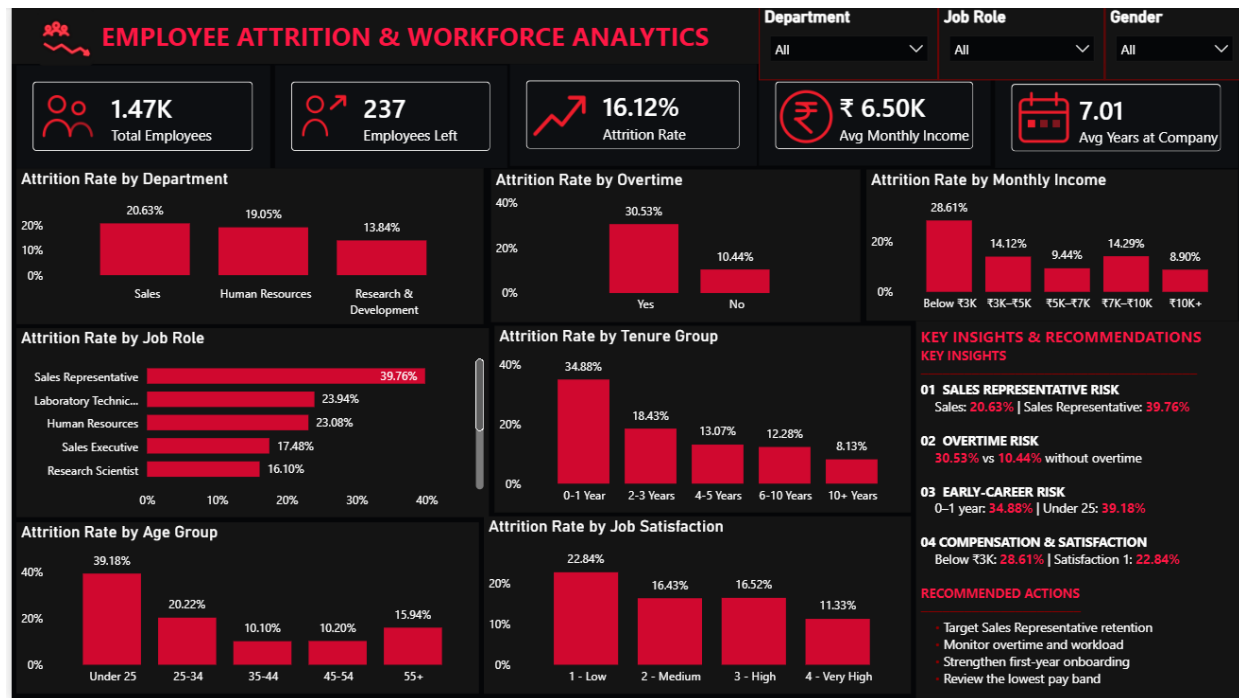
Dashboard Layout

The page is organized into four clear zones:

- Top — dashboard title and the three slicers
- Second section — the five KPI cards, so the stakeholder immediately sees overall workforce condition
- Main analytical area — the seven attrition-rate visuals (Department, Job Role, Overtime, Monthly Income, Tenure, Age, Job Satisfaction)
- Right-side panel — Key Insights & Recommendations

Rather than adding a second page or cluttering the dashboard with paragraphs of text, insights and recommendations were kept in a dedicated side panel. This lets the dashboard itself answer “what is happening?” while the panel answers “what does it mean, and what should HR do about it?”

Dashboard View



Final Power BI dashboard — Dark + Neon Red theme, showing KPI cards, slicers, the seven attrition-rate visuals, and the Key Insights & Recommendations panel.

KPI Analysis

Five KPI cards give HR an immediate, at-a-glance read on workforce health before a single chart is examined. With no slicers applied, the default values are:

KPI	Value
Total Employees	1.47K
Employees Left	237
Attrition Rate	16.12%
Average Monthly Income	₹6.50K
Average Years at Company	7.01

Visual Analysis

The dashboard began with around a dozen charts; redundant ones were trimmed so the final set focuses almost entirely on attrition rate — the metric that actually matters to the business question. Each visual, its finding, and its business meaning are summarized below.

Visual	Finding	Business Meaning
Department	Sales: 20.63%	Highest departmental attrition
Job Role	Sales Representative: 39.76%	Risk concentrated in one role
Overtime	30.53% vs 10.44%	Strong overtime-related risk signal
Tenure	0–1 year: 34.88%	Early-tenure retention concern
Age	Under 25: 39.18%	Early-career risk
Income	Below ₹3K: 28.61%	Highest income-band attrition
Satisfaction	Low: 22.84%	Lower satisfaction associated with higher attrition

Interactive Features

Three slicers were added to the dashboard:

- Department — lets the user focus on one department at a time
- Job Role — lets the user drill into a specific role
- Gender — enables demographic comparison

Whenever a slicer selection changes, every KPI card and every analytical visual on the page updates through Power BI's filter context — this is what makes it a dashboard rather than a static report.

Key Insights

01 — Sales Representative Risk

Sales as a department sits at 20.63% attrition, but Sales Representatives specifically reach 39.76% — the risk is concentrated in that one role.

02 — Overtime Risk

Employees working overtime have a 30.53% attrition rate compared with 10.44% for those who don't.

03 — Early-Career Risk

Employees in their first year have a 34.88% attrition rate, and employees under 25 reach 39.18%.

04 — Compensation & Satisfaction

Employees earning below ₹3K show 28.61% attrition, while the lowest satisfaction band sits at 22.84%.

Recommendations

1. Target Sales Representative retention

Look specifically at role expectations, incentives, workload, management, and career progression for this role — rather than rolling out a Sales-department-wide intervention.

2. Monitor overtime and workload

Review overtime patterns, workload distribution, and staffing levels, paying particular attention to cases where overtime overlaps with other risk factors.

3. Strengthen first-year onboarding

Introduce 30/60/90-day check-ins, mentoring, clearer role definitions, and early-career development paths.

4. Review the lowest compensation band

Investigate the below-₹3K group specifically and consider targeted retention or compensation actions rather than an across-the-board raise.

Testing & Validation

Building the dashboard was not the last step — it was also tested before being considered finished.

Slicers

Department, Job Role, and Gender were checked individually, confirming that selecting a category correctly updates the relevant KPIs and visuals.

DAX

Every KPI and attrition calculation was re-checked against the underlying data to make sure the numbers held up.

Formatting

Percentages, currency formatting, year values, labels, and chart titles were reviewed for consistency and visual quality.

Issues Found and Fixed

- Age Group ordering
- Income Group ordering
- Job Satisfaction labels
- Job Role scrollbar behavior
- KPI label truncation
- Visual spacing
- Dashboard colors

The final stage of the project was not just “make it pretty” — it also verified that the dashboard actually works.

Business Value

By breaking a single 16.12% attrition rate down into department, role, tenure, age, income, and satisfaction segments, the dashboard gives HR a way to prioritize retention effort where it will have the most impact, rather than applying broad, undifferentiated interventions. Because it is built on live DAX measures and interactive

slicers rather than a fixed set of static charts, HR can continue to explore new questions — by department, role, or gender — without needing a new report built each time.

Limitations

One principle was maintained carefully throughout the analysis: overtime is not claimed to cause attrition — only to be associated with higher attrition. The same applies to income: lower income is associated with higher attrition, not proven to cause it. This distinction matters because the dashboard is descriptive and diagnostic analysis, not a causal study or a machine-learning prediction model. It is a small thing, but it is worth calling out explicitly in documentation — it reflects the level of analytical rigor the project was held to.

Future Enhancements

Building on the current dashboard, natural next steps include:

- Extending the model with a predictive attrition-risk score, built on top of the existing descriptive segments
- Adding a time dimension, if historical snapshots become available, to track whether attrition in each segment is rising or falling
- Expanding the slicer set (for example, by tenure or age group) for even finer-grained drill-down

Conclusion

In one sentence: raw employee-level HR data was transformed into an interactive Power BI dashboard that identifies where attrition is concentrated, highlights the workforce factors associated with higher attrition, and translates those findings into targeted HR retention actions.

RAW EMPLOYEE DATA → DATA PREPARATION → MEANINGFUL GROUPS (Age / Income / Tenure / Satisfaction) → DAX MEASURES → KPI ANALYSIS → ATTRITION ANALYSIS → INTERACTIVE SLICERS → HIGH-RISK SEGMENTS → BUSINESS INSIGHTS → RECOMMENDATIONS → EXECUTIVE DASHBOARD